UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

STAT 576: Empirical Process Theory and Weak Convergence Fall 2025
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Stein's Method

Content.

1. Solving Stein's ODE
2. Smoothness of solutions to Stein's ODE
3. Stein's method for proving ordinary CLT in Wasserstein metric

Notation: Let W be an arbitrary random variable and $Z \sim N(0,1)$. Recall the 'overarching' theme: our goal is to bound

$$\sup_{g \in \mathcal{D}} |\mathbb{E}[g(W)] - \mathbb{E}[g(Z)]| .$$

Stein's Idea: Recall that Stein proved:

$$W \overset{d}{=} Z \quad \iff \quad \mathbb{E}[A(f(W))] = 0 \quad \forall f \text{ absolutely continuous, } f : \mathbb{R} \to \mathbb{R},$$

with $\mathbb{E}|A(f(W))| < \infty$.

The Stein operator is

$$A(f(x)) = f'(x) - xf(x).$$

Intuition: If $W \approx Z$, then $\mathbb{E}[Af(W)]$ should be close to zero.

Lemma 1. *Assume that there exists a class $\mathcal{D}'$ of functions such that for every $g \in \mathcal{D}$, there exists $f \in \mathcal{D}'$ with*

$$g(x) - \mathbb{E}g(Z) = f'(x) - xf(x) = A(f)(x).$$

Then;

$$\sup_{g\in\mathcal{D}} \big|\mathbb{E}[g(W)] - \mathbb{E}[g(Z)]\big| \;\leq\; \sup_{f\in\mathcal{D}'} \big|\mathbb{E}[f'(W)] - \mathbb{E}[Wf(W)]\big|.$$

Remarks

- Given $\mathcal{D}$, it makes sense to choose $\mathcal{D}'$ to be as small as possible.

Road Map

1. Solve Stein's ODE.
2. Use the solution to pinpoint properties of $\mathcal{D}'$.

Lemma 2. *Given $g : \mathbb{R} \to \mathbb{R}$, $\mathbb{E}|g(Z)| < \infty$,*

$$f(x) = e^{x^2/2} \int_{-\infty}^{x} \big(g(y) - \mathbb{E}g(Z)\big) e^{-y^2/2}\, dy$$

is an absolutely continuous solution to Stein's ODE. Moreover, it is the unique solution with

$$\lim_{|x|\to\infty} f(x)e^{-x^2/2} = 0.$$

Proof

$$\begin{aligned}
\frac{d}{dx}\big(e^{-x^2/2} f(x)\big) &= e^{-x^2/2} f'(x) - x e^{-x^2/2} f(x). \\
&= e^{-x^2/2}\big(f'(x) - xf(x)\big) \\
&= e^{-x^2/2}\big(g(x) - \mathbb{E}g(Z)\big)
\end{aligned}$$

Integrating, f is easily seen to be a solution of Stein's ODE:

Given g, let $\bar{f}$ be another solution to Stein's ODE. Then

$$\frac{d}{dx}\left(e^{-x^2/2}f(x)\right) = \frac{d}{dx}\left(e^{-x^2/2}\bar{f}(x)\right)$$
$$\Rightarrow e^{-x^2/2}\big(f(x) - \bar{f}(x)\big) = 0.$$

Hence,

$$\bar{f}(x) = f(x) + ce^{x^2/2}, \quad c \in \mathbb{R}.$$

So any solution $\bar{f}$ is of the form above for some constant c.

Explicit Formula

$$f(x) = e^{x^2/2}\int_{-\infty}^{x}\big(g(y) - \mathbb{E}g(Z)\big)e^{-y^2/2}\,dy.$$

Moreover,

$$\lim_{x\to\infty} f(x)e^{-x^2/2} = 0$$

by the dominated convergence theorem (since $\int_{-\infty}^{\infty} g(y)e^{-y^2/2}\,dy = \mathbb{E}g(Z)$), as justified below:

Pointwise convergence as $x \to \infty$ holds since

$$g(y)e^{-y^2/2}\mathbf{1}_{\{y\le x\}} \to g(y)e^{-y^2/2}.$$

Moreover,

$$|g(y)|e^{-y^2/2}\mathbf{1}_{\{y\le x\}} \le |g(y)|e^{-y^2/2}$$

where the latter is integrable by assumption. Thus, the conditions of DCT are satisfied.

Lemma 3. *Let $g : \mathbb{R} \to \mathbb{R}$ be Lipschitz. Then*

$$f(x) = \int_0^1 \frac{-1}{2\sqrt{t(1-t)}}\,\mathbb{E}\Big[Zg\big(\sqrt{t}\,x + \sqrt{1-t}\,Z\big)\Big]\,dt$$

solves Stein's ODE. In fact, it is the same as in the previous lemma.

Proof. If g is C-Lipschitz, then $\|g'\|_\infty \le C$.

$$f'(x) = \int_0^1 \frac{-1}{2\sqrt{t(1-t)}}\sqrt{t}\,\mathbb{E}\Big[Zg'\big(\sqrt{t}\,x + \sqrt{1-t}\,Z\big)\Big]\,dt$$

Automatically:

$$|f'(x)| \le \left| \int_0^1 \frac{1}{2\sqrt{1-t}} \, \mathbb{E}\big[|Z| \, \|g''\|_\infty\big] \, dt \right|$$

$$\le \|g'\|_\infty \, \mathbb{E}|Z| \left| \int_0^1 \frac{1}{2\sqrt{1-t}} dt \right| = \sqrt{\tfrac{\pi}{2}} \, \|g'\|_\infty.$$

□

Lemma 4 (Stein's Lemma)**.**

$$\mathbb{E}[Z\phi(Z)] = \mathbb{E}\big[\phi'(Z)\big] \, .$$

Take $\phi(z) = g(\sqrt{t}\, x + \sqrt{1-t}\, z)$*. Then*

$$\mathbb{E}\Big[Z g\big(\sqrt{t}\, x + \sqrt{1-t}\, Z\big)\Big] = \sqrt{1-t}\mathbb{E}\Big[\, g'\big(\sqrt{t}\, x + \sqrt{1-t}\, Z\big)\Big] \, .$$

Thus,

$$f'(x) - xf(x) = \int_0^1 \mathbb{E}\left[\left(\frac{-Z}{2\sqrt{t(1-t)}} + \frac{x}{2\sqrt{t}} \right) g'\big(\sqrt{t}\, x + \sqrt{1-t}\, Z\big) \right] dt$$

$$= \int_0^1 \mathbb{E}\left[\frac{d}{dt} g'\big(\sqrt{t}\, x + \sqrt{1-t}\, Z\big) \right] dt = g(x) - \mathbb{E}g(Z).$$

Alternative Representation

$$f(x) = -\int_0^1 \frac{1}{2\sqrt{t}} \, \mathbb{E}\Big[g'\big(\sqrt{t}\, x + \sqrt{1-t}\, Z\big)\Big] \, dt, \quad \text{(using Stein's identity).}$$

Since

$$\mathbb{E}\Big[Z g(\sqrt{t}\, x + \sqrt{1-t}\, Z)\Big] = \sqrt{1-t}\, \mathbb{E}\Big[g'\big(\sqrt{t}\, x + \sqrt{1-t}\, Z\big)\Big] \, ,$$

we obtain

$$|f(x)| \le \int_0^1 \frac{1}{2\sqrt{t}} \, \mathbb{E}\big[\|g''\|_\infty\big] \, dt.$$

$$\le \|g''\|_\infty \int_0^1 \frac{1}{2\sqrt{t}} \, dt = \|g''\|_\infty.$$

Hence,

$$\|f\|_\infty \le \|g''\|_\infty.$$

Lemma 5. *Given a bounded $g : \mathbb{R} \to \mathbb{R}$, there exists an absolutely continuous $f : \mathbb{R} \to \mathbb{R}$ solving Stein's ODE*

$$f'(x) - xf(x) = g(x) - \mathbb{E}g(Z),$$

while also satisfying:

1. $\|f\|_\infty \leq \sqrt{\pi/2}\,\|g - Ng\|_\infty$
2. $\|f'\|_\infty \leq 2\,\|g - Ng\|_\infty$, *where* $Ng = \mathbb{E}g(Z)$

If g is Lipschitz (i.e. not necessarily bounded), we can take:

3. $\|f\|_\infty \leq \|g\|_\infty$
4. $\|f\|_\infty \leq \sqrt{\pi/2}\,\|g'\|_\infty$
5. $\|f''\|_\infty \leq 2\,\|g''\|_\infty$

Corollary 1.

$$d_{Was}(W, Z) := \sup_{g:\, g \text{ is 1-Lip}} \big|\mathbb{E}g(W) - \mathbb{E}g(Z)\big|$$

$$\leq \sup_{f \in \mathcal{D}'} \big|\mathbb{E}f'(W) - \mathbb{E}[Wf(W)]\big|,$$

where

$$\mathcal{D}' = \left\{ f : \mathbb{R} \to \mathbb{R} \;\middle|\; \begin{array}{l} (a)\ \|f\|_\infty \leq 1 \\ (b)\ \|f'\|_\infty \leq \sqrt{2/\pi} \\ (c)\ \|f''\|_\infty \leq 2 \end{array} \right\}.$$

Theorem 1. *Let X_i's be independent, zero mean, unit variance with $\mathbb{E}|X_i|^3 < \infty$. Then*

$$d_{Was}\left(\frac{1}{\sqrt{n}}\sum_{j=1}^{n} X_j, Z\right) \;\leq\; \frac{3}{n^{3/2}}\sum_{j=1}^{n}\mathbb{E}|X_j|^3.$$

Remark 1. *This result, as is, is incomparable with*

$$d_{KS}\left(\tfrac{1}{\sqrt{n}}S, Z\right) = O\left(\tfrac{1}{\sqrt{n}}\right).$$

Indeed,

$$d_{KS}(\cdot, Z) \leq \sqrt{2c d_{Was}(\cdot, Z)}$$

if the density of Z is upper bounded by c.

So the given rate of $O(n^{-1/2})$ for Kolmogorov–Smirnov distance is suboptimal.

Proof. Let $S = \sum_i X_i$, $W = \frac{1}{\sqrt{n}} S$. Define

$$W_i := W - \tfrac{1}{\sqrt{n}} X_i.$$

We will bound

$$\sup_{f \in \mathcal{D}} \left| \mathbb{E} f'(W) - \mathbb{E}[W f(W)] \right|$$

with $\mathcal{D}$ the same as in the corollary.

Note that since $W_i \perp X_i$ we have that $\mathbb{E}[X_i f(W_i)] = 0$.

$$W f(W) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} X_i f(W).$$

Now expand $f(W)$:

$$f(W) = f\Big(W_i + \tfrac{1}{\sqrt{n}} X_i\Big) = f(W_i) + \tfrac{1}{\sqrt{n}} X_i f'(W_i) + \frac{X_i^2}{2n} f''(\theta_i),$$

for some $\theta_i \in (W_i, W)$.

Thus,

$$W f(W) = \frac{1}{\sqrt{n}} \sum_i X_i \Big[f(W_i) + \tfrac{1}{\sqrt{n}} X_i f'(W_i) + \tfrac{X_i^2}{2n} f''(\theta_i) \Big].$$

Taking expectations:

$$\mathbb{E}[W f(W)] = \frac{1}{n} \sum_i \mathbb{E}\big[X_i^2 f'(W_i)\big] + \frac{1}{2n\sqrt{n}} \sum_i \mathbb{E}\big[X_i^3 f''(\theta_i)\big] \,.$$

Now,

$$\begin{aligned} \left| \mathbb{E} f'(W) - \mathbb{E}[W f(W)] \right| &\leq \left| \mathbb{E} f'(W) - \tfrac{1}{n} \sum_i \mathbb{E} f'(W_i) - \frac{1}{2n\sqrt{n}} \sum_i \mathbb{E} X_i^3 \, f''(\theta_i) \right| \\ &\leq \left| \mathbb{E} f'(W) - \tfrac{1}{n} \sum_i \mathbb{E} f'(W_i) \right| + \frac{1}{2n\sqrt{n}} \underbrace{\sum_i \mathbb{E}|X_i|^3}_{\frac{2}{n^{3/2}} \sum_i |X_i|^3} \underbrace{\|f''\|_\infty}_{\leq 2} \end{aligned}$$

Bounding the final term

$$\Big|\frac{1}{n}\sum_i \mathbb{E}f'(W) - \mathbb{E}f'(W_i)\Big| \le \frac{1}{n}\sum_i \Big|\mathbb{E}f'(W) - \mathbb{E}\big[f'(W_i)\big]\Big|$$

Now expand:

$$f'(W)-f'(W_i) = f'(W_i+\tfrac{1}{\sqrt{n}}X_i)-f'(W_i) = \frac{1}{\sqrt{n}}X_i f''(\bar{\theta}_i), \quad \bar{\theta}_i \in (W_i, W).$$

Thus,

$$|f'(W) - f'(W_i)| \le \frac{1}{\sqrt{n}}|X_i|\,\|f''\|_\infty.$$

Taking expectations:

$$|\mathbb{E}f'(W) - \mathbb{E}f'(W_i)| \le \frac{1}{\sqrt{n}}\,\mathbb{E}|X_i|\,\|f''\|_\infty.$$

Therefore, we arrive at:

$$\mathbb{E}f'(W) - \mathbb{E}Wf(W) \le \frac{2}{n^{3/2}}\sum \mathbb{E}|X_i| + \frac{1}{n^{3/2}}\mathbb{E}|X_i|^3$$

Claim 1.

$$\mathbb{E}|X_i| \le \mathbb{E}|X_i|^3.$$

To see the claim: by Jensen,

$$(\mathbb{E}|X_i|^3)^{1/3} \ge (\mathbb{E}|X_i|^2)^{1/2}.$$

Since $\mathbb{E}|X_i|^2 = 1$, it follows

$$(\mathbb{E}|X_i|^3)^{1/3} \ge 1 \quad \Rightarrow \quad (\mathbb{E}|X_i|^3)^{1/3} \ge \mathbb{E}|X_i|.$$

Also, trivially as $\mathbb{E}|X_i|^3 \ge 1$,

$$(\mathbb{E}|X_i|^3)^{1/3} \le \mathbb{E}|X_i|^3.$$

And finally we arrive at the following bound:

$$|\mathbb{E}f'(W) - \mathbb{E}[Wf(W)]| \le \frac{3}{n^{3/2}}\sum_i \mathbb{E}|X_i|^3.$$

Taking $\sup_{f\in\mathcal{D}}$ finishes the proof.

□